

■ Transpose

`Transpose[list]` transposes the first two levels in *list*.

`Transpose[list, { $n_1$ ,  $n_2$ , ...}]` transposes *list* so that the n_k^{th} level in *list* is the k^{th} level in the result.

Example: `Transpose[{{a,b},{c,d}}] → {{a, c}, {b, d}}`. ■ `Transpose` gives the usual transpose of a matrix.

■ Acting on a tensor $T_{i_1 i_2 i_3 \dots}$ `Transpose` gives the tensor $T_{i_2 i_1 i_3 \dots}$. ■ `Transpose[list, { $n_1$ ,  $n_2$ , ...}]` gives the tensor $T_{i_{n_1} i_{n_2} \dots}$. ■ The level specifications n_k do not necessarily have to be distinct. ■ See page 446. ■ See also: `Flatten`.